MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

The population density of Iceland, in people per square kilometer of land area, increased from 2.5 in 1990 to 3.3 in 2014. During this time period, the land area of Iceland was 100,250 square kilometers. By how many people did Iceland’s population increase from 1990 to 2014?

A. 330,825

B. 132,330

C. 125,312

D. 80,200

Q3 PROBLEM-SOLVING AND DATA ANALYSIS

At a particular track meet, the ratio of coaches to athletes is 1 to 26. If there are x coaches at the track meet, which of the following expressions represents the number of athletes at the track meet?

A. $\frac{x}{26}$

B. $26x$

C. $x + 26$

D. $\frac{26}{x}$

Q4

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 19.700 miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg? 1 mile $= 1{,}760$ yards

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 PROBLEM-SOLVING AND DATA ANALYSIS

The International Space Station orbits Earth at an average speed of 4.76 miles per second. What is the space station’s average speed in miles per hour?

A. 285.6

B. 571.2

C. 856.8

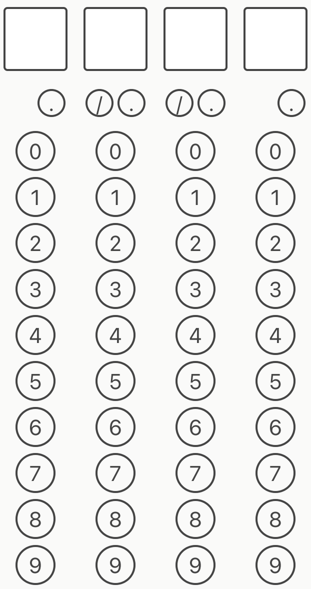
D. 17,136.0

Q6

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

One of a planet's moons orbits the planet every 252 days. A second moon orbits the planet every 287 days. How many more days does it take the second moon to orbit the planet 29 times than it takes the first moon to orbit the planet 29 times?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

How many tablespoons are equivalent to 14 teaspoons? $3 \text{ teaspoons} = 1 \text{ tablespoon}$

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

A distance of 61 furlongs is equivalent to how many feet?

$1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 PROBLEM-SOLVING AND DATA ANALYSIS

$$d = 55t$$

The equation above can be used to calculate the distance d, in miles, traveled by a car moving at a speed of 55 miles per hour over a period of t hours. For any positive constant k, the distance the car would have traveled after $9k$ hours is how many times the distance the car would have traveled after $3k$ hours?

A. 3

B. 6

C. $3k$

D. $6k$

Q10

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

On April 18, 1775, Paul Revere set off on his midnight ride from Charlestown to Lexington. If he had ridden straight to Lexington without stopping, he would have traveled 11 miles in 26 minutes. In such a ride, what would the average speed of his horse have been, to the nearest tenth of a mile per <u>hour</u>?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.